Discovery Breakthrough Profile for `p-vs-np`

Neuro-symbolic AI Group
Idiap Research Institute

March 18, 2026

Abstract

This report classifies the blocked discovery frontier by the conceptual breakthrough required to make progress. It reframes the frontier from missing theorems to breakthrough families such as realization, transport, alignment, explicitness, constructive instantiation, and semantic bridging.

1 Summary

- The discovery layer should stop restating blocked lemmas as if they were already search strategies.
- Most frontier pressure is now in bridge-layer breakthroughs: realization, transport, alignment, explicitness, and semantic bridging.
- Target-grounded routes should be treated as activation-theorem searches, not as target-theorem discovery.
- The highest-pressure breakthrough family is 'abstraction-search' via obstruction class 'mcsp-magnification'.

2 mcsp-magnification

Breakthrough family. abstraction-search

Pressure score. 5.818

Required move. Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.

Abstraction required. A theorem-family-specific carrier, invariant, or bridge law that adds new mathematical content.

Minimal theorem shape. State a route-specific theorem that would resolve the blocked interface and make the next proof obligation materially sharper.

Major subresult type. exploratory-conjecture

Search Modes

- typed abstraction search
- bridge split
- counterfactual sharpening

Anti-Patterns

- repeating the same blocked statement without new typed structure

3 carrier-realization

Breakthrough family. realization

Pressure score. 5.818

Required move. Turn a typed but schematic carrier into an explicit mathematical object family with admissibility conditions and a forcing theorem.

Abstraction required. An explicit obstruction family, witness family, surrogate promise problem, or quantitative measure that is realized from the source theorem rather than only named by role.

Minimal theorem shape. If source theorem S holds in regime R, then one can construct an explicit carrier C with admissibility conditions A such that C preserves invariant I and supports a route-specific forcing lemma to target T.

Major subresult type. bridge object/tool

Search Modes

- explicit carrier construction
- equivalence search between live carrier encodings
- surrogate problem design with forcing semantics

Anti-Patterns

- renaming the carrier without adding realizability conditions
- adding more obligations without exhibiting a concrete carrier family

4 formula-size-transport

Breakthrough family. transport

Pressure score. 5.060

Required move. Find a theorem that preserves the relevant invariant across a cross-model move into formula-size.

Abstraction required. A transport law, compilation theorem, or complexity measure that is stable under the source-to-target model shift.

Minimal theorem shape. For every admissible carrier C in the nonuniform-circuit regime, there is a transformed carrier C' in the formula-size regime preserving invariant I and retaining enough quantitative strength to imply target T.

Major subresult type. transfer principle

Search Modes

- cross-model transport principle
- invariant-preserving compilation theorem
- representation normal form for formula-size transfer

Anti-Patterns

- assuming the target theorem already provides the transport
- strengthening the source lower bound without addressing the model shift

5 target-grounding

Breakthrough family. bridge-specificity

Pressure score. 5.014

Required move. Stop treating the target theorem as the open problem and relocate search to the bridge that would activate it.

Abstraction required. No new target theorem; discovery should instead seek the carrier, transfer law, or alignment theorem that makes the target theorem applicable.

Minimal theorem shape. Target theorem T is fixed. The real theorem under search is the local activation theorem A that verifies T's hypotheses from current artifacts.

Major subresult type. research recommendation

Search Modes

- activation theorem search
- hypothesis verification search
- bridge-localization refinement

Anti-Patterns

- rediscovering the target theorem
- claiming novelty at the target instead of at the activation bridge

6 local-reduction-barrier

Breakthrough family. barrier-aware transfer

Pressure score. 1.984

Required move. Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.

Abstraction required. A barrier-aware reduction gadget family or an impossibility theorem explaining why the reduction regime cannot be crossed.

Minimal theorem shape. Either construct a local reduction family G satisfying locality bounds L and preserving witness structure W, or prove that any such G would contradict barrier theorem B.

Major subresult type. barrier result

Search Modes

- barrier-aware gadget design
- impossibility theorem search
- reduction regime decomposition

Anti-Patterns

- ignoring the known local-reduction barrier theorem
- using generic NP-hardness language without locality constraints

7 target-alignment

Breakthrough family. alignment

Pressure score. 1.897

Required move. Identify or prove an equivalence between the current carrier class and the object class required by the target theorem.

Abstraction required. An alignment theorem, correspondence theorem, or mediating normal form linking the local carrier to the target's sparse-problem or distinguisher objects.

Minimal theorem shape. Every admissible carrier C from the local route induces an object D in the target theorem's class, and D preserves the quantitative property needed to activate the target theorem.

Major subresult type. transfer principle

Search Modes

- object-class correspondence search
- normal-form mediation
- dual characterization of the target object

Anti-Patterns

- treating topical similarity as object equivalence
- adding a stronger carrier without specifying the target object class map

8 general-magnification

Breakthrough family. alignment

Pressure score. 1.897

Required move. Map a problem-specific carrier into a general magnification framework without losing the quantitative threshold that matters.

Abstraction required. A general-purpose sparse-problem or distinguisher formulation that faithfully receives the local carrier.

Minimal theorem shape. Problem-specific object C induces general object G with preserved sparsity or approximation parameters sufficient to trigger the general magnification theorem.

Major subresult type. transfer principle

Search Modes

- generalization by faithful embedding
- parameter-preserving reformulation
- problem-specific to generic theorem bridge

Anti-Patterns

- moving to the generic theorem without a preserved parameter map
- using the general theorem only as motivation rather than as a typed target

9 pit-upgrade

Breakthrough family. explicitness

Pressure score. 1.752

Required move. Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.

Abstraction required. An explicit generator, certificate family, or algebraic witness system that is constructive enough for PIT rather than merely semantic.

Minimal theorem shape. If semantic object C exists for the relevant circuit class, then one can uniformly and efficiently extract an explicit structure G that yields deterministic PIT for that class.

Major subresult type. bridge object/tool

Search Modes

- explicit extractor or generator search
- certificate map construction
- semantic-to-algorithmic upgrade theorem

Anti-Patterns

- equating a black-box hitting-set statement with an explicit generator
- rephrasing derandomization as semantics without uniformity or efficiency

10 schema-to-examples

Breakthrough family. constructive instantiation

Pressure score. 1.641

Required move. Turn an abstract schema into explicit natural example families or prove that the schema cannot instantiate naturally in the current way.

Abstraction required. A construction template, example-generating reduction family, or a canonicalization principle for natural examples.

Minimal theorem shape. From abstract schema S, construct an explicit family of natural instances E_n satisfying properties P and realizing target theorem T.

Major subresult type. standalone theorem

Search Modes

- canonical example construction
- template instantiation
- example-preserving reduction search

Anti-Patterns

- restating the schema as if it already supplied examples
- listing examples without a constructive route from the local abstraction

11 characterization-bridge

Breakthrough family. semantic bridge

Pressure score. 1.621

Required move. Bridge a local definitional object to a global characterization theorem.

Abstraction required. A semantic correspondence, duality, witness extraction principle, or completeness notion that turns the local definition into the target biconditional.

Minimal theorem shape. Local definitional object D has property P if and only if global characterization condition Q holds, with both directions proved at the right level of abstraction.

Major subresult type. transfer principle

Search Modes

- semantic correspondence theorem
- duality or normal form search
- two-way implication decomposition

Anti-Patterns

- normalizing syntax without adding semantic force
- treating a definition as if it already contained the characterization

12 explicitness-gap

Breakthrough family. explicitness

Pressure score. 1.342

Required move. Add uniform, explicit, efficiently constructible content to an otherwise semantic object.

Abstraction required. A uniform construction rule, extractor, or explicit family indexed by input length and class parameters.

Minimal theorem shape. For every length parameter n, the existential object C_n can be constructed or certified explicitly in time class X while preserving property P.

Major subresult type. bridge object/tool

Search Modes

- uniformization
- effective construction
- certificate extraction

Anti-Patterns

- accepting existence without algorithmic access
- shifting to a broader semantic object that is still non-explicit